Nalezinková-Kellner-Svaj#ík

sarka

Table of Contents

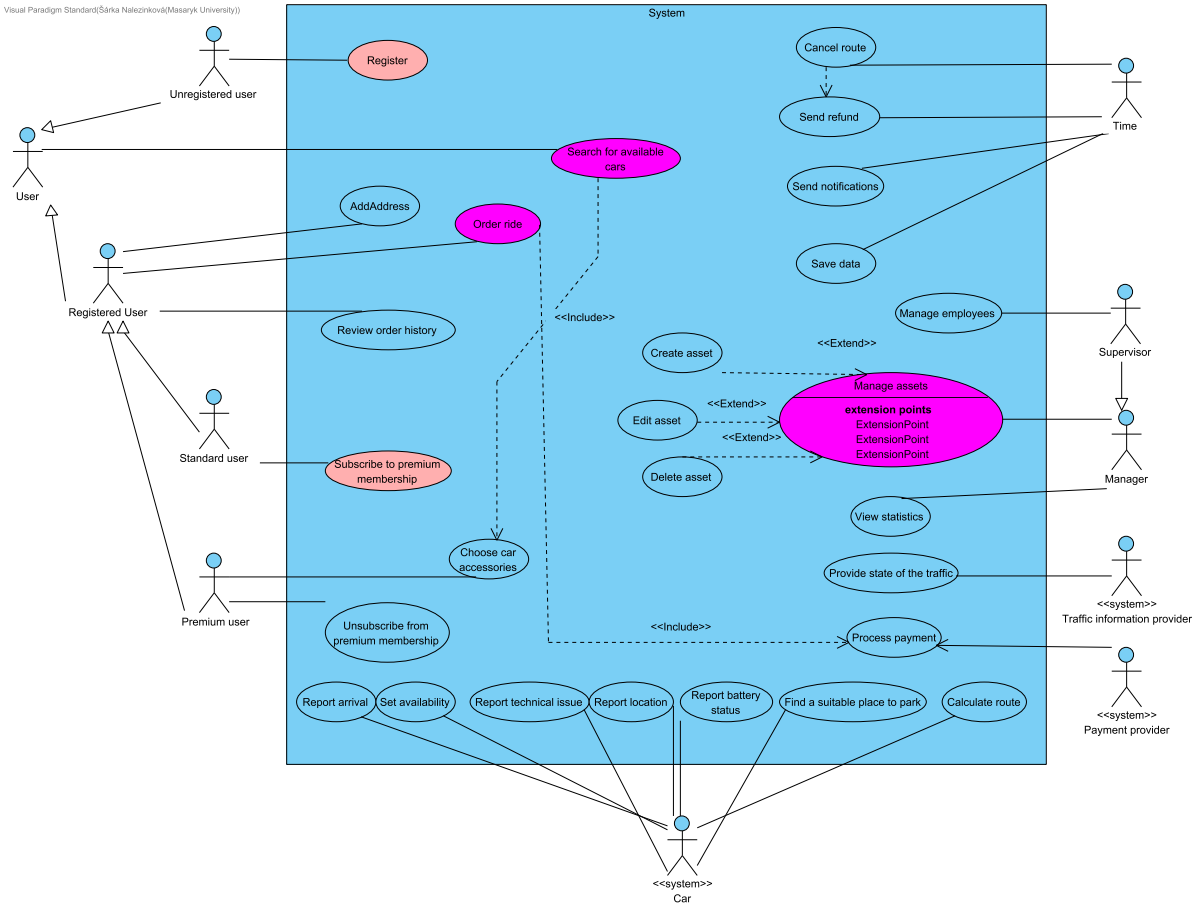
Use Case Diagram1....4
Class Diagram1....16
Register....19
Subscribe to premium membership....22
Entity Relationship Diagram1....24
Porušení 2 NF....26
Porušení 3 NF....28

Table of Figures

Use Case Diagram1....4
Class Diagram1....16
Register....19
Subscribe to premium membership....22
Entity Relationship Diagram1....24
Porušení 2 NF....26
Porušení 3 NF....28

Use Case Diagram

Use Case Diagram1



Summary

Name	Description
User	
Registered User	
Unregistered user	
Standard user	
Premium user	
Car	
Supervisor	
Time	
Manager	
Traffic information provider	
Payment provider	
Manage employees	
AddAddress	
System	Functional requirements: 1. Customer searches for available cars

	2. Unregistered customer registers 3. Registered customer orders a ride 4. Registered customer cancels an ordered ride 5. Registered customer views order history 6. Standard customer subscribes to monthly premium membership 7. Standard customer subscribes to yearly premium membership 8. System changes the Standard status to Premium upon subscription 9. Premium customer chooses required car accessories 10. Premium customer has a 20% discount on all orders 11. Manager creates a new asset 12. Manager edits an asset 13. Manager deletes an asset 14. Manager views statistics 15. Car reports arrival 16. Car sets its availability 17. Car reports a technical issue 18. Car sends its location 19. Car reports battery status 20. System sends notifications 21. System calculates route 22. System sends route 23. System saves data 24. System cancels route 25. System sends cancellation fee 26. System finds suitable place to park 27. Payment provider processes payment 28. Traffic information provider provides the state of the traffic Non-functional requirements: 1. System will process received location within 1 second 2. Car is sending its position to the system every second 3. Car waits for clients for 15 minutes 4. The system immediately notifies the system operator when a problem occurs during a route
Manage assets	Description: Manager gets the option to create, edit, or delete assets. Primary actors: Manager

	Secondary actors: -
	Preconditions: -
	Main flow:
	Manager opens the asset management section of the system
	Manager chooses to create new asset or selects an asset to edit/delete
	if Manager chooses to create new asset{
	Create asset()}
	if Manager selects an existing asset{
	if Manager chooses to edit the asset{
	Edit asset()}
	if Manager chooses to delete the asset{
	if Manager confirms the choice{
	Delete asset()}
	else{
	back to home screen}}
	System saves the updated assets}
	Post-conditions:
	One of the extensions is called
	Alternative flow:
	Manager logs out:
	Action gets canceled

Unsubscribe from premium membership	A premium user can cancel their premium subscription. The system updates the user's membership status and deletes their access to premium features.
Provide state of the traffic	The system obtains traffic information from an external traffic information provider. It uses this information to plan the routes of the cars.
Review order history	Registered users can view their previous orders and rides. The system displays relevant historical information, such as booking details and ride records.
Find a suitable place to park	When the car isn't carrying passengers and is available, the system finds a place to park and navigates the car there.
Process payment	The system processes payment for a ride or a premium membership. It communicates with the external payment provider to handle the transaction and determine its outcome.
Send refund	The system sends back the money for the canceled order after deducting a cancellation fee.

Cancel route	If the passengers don't get onboard the vehicle within 15 minutes after it arrives to the pickup point, the system cancels the order.
Save data	The system saves all data (orders, car information, routes, users, etc.) every second.
Calculate route	All routes to the destinations are computed by the system based on the location data from the car and then sent to the vehicle. The route is computed based on the current position of the vehicle and the traffic situation. The route is represented by a collection of waypoints that the autonomous car is able to follow.
Send notifications	The system sends notifications to relevant users about the status of their route, such as changes of the route, cancellations, or vehicle arrival.
Choose car accessories	During search, premium users can choose additional car accessories, such as entertainment systems, comfort accessories, etc.
Report battery status	The car sends its battery status to the system every second. Whenever the battery charge drops below 40%, and the vehicle is not carrying out any order at the moment, the system will navigate the vehicle to the nearest free charging station.
Report location	The vehicle sends its position to the system every second.
Report technical issue	The car reports a technical problem or malfunction. In response, the system modifies the route or finds a suitable place to park if needed.
Set availability	The car sets its status as available when it isn't currently driving passengers or en route to a pickup location or when its battery is above 40% and as unavailable otherwise.
Report arrival	The car's system reports that it has arrived at the pickup location. The system records the arrival information so that the route status can be updated.
View statistics	The manager can access statistical information about system operations, users, assets and their usage.
Delete asset	The manager removes an asset that is no longer needed or available. The system updates its records to reflect the removal.
Edit asset	The manager modifies the details of an existing asset. The system saves the changes to ensure the asset information remains current.

Create asset	The manager adds a new asset to the system by entering its relevant details. The new asset is saved into the system.
Subscribe to premium membership	User subscribes to premium membership using one of the approved payment methods, the system applies a discount if it applies checks if the payment was successful and sets the users account status to premium.

Order ride	Description:
	The Registered User orders a car by selecting an available ride from the search results and completing the payment process.
	Primary actors: Registered User
	Secondary actors: Payment provider
	Preconditions: The user searched for a ride and at least one ride was provided
	Main flow:
	Registered User chooses a ride from the list of available rides
	Registered User is asked to confirm the selected ride
	if Registered User cancels the selection{
	Order ride()}
	else{
	if User's card details aren't saved in the system{
	Registered User is asked to fill in card information}
	else{
	Payment provider Processes payment
	if payment is successful{
	Car sets availability to unavailable
	System calculates route
	System sends route to car}}}
	Post-conditions:
	Car is unavailable, Firm received money
	Alternative flow:
	Payment fails:
	User receives an option to retry payment or cancel ride
	User logs out before payment:
	Action gets canceled

Search for available cars	Description: The user searches for a suitable car or ride from point A to point B, with the option to select additional accessories for premium users. Primary actors: User Secondary actors: -

	Preconditions: - Main flow: User clicks the button to search for a ride User inputs pickup point, destination, pickup time, number of passengers if User is Premium User{ User gets the option to input wanted equipment} System validates all fields System searches available rides that fit the User's requirements System creates a route from pickup to destination System displays available rides to the User Post-conditions: Available suitable rides are displayed to the user Alternative flow: No cars are available: User gets the option to change their requirements User logs out: Action gets canceled No route from pickup point to destination has been found by the system: User gets the option to change their requirements
Register	User submits username, email, password, and payment card details on the registration page, the system validates the email format, password requirements, and payment card validity, checks if the email is already registered, sends a confirmation link to the user, verifies if the link is valid and confirmed, hashes the password, creates the user account in the database, and sends a welcome email.

Details

User
Registered User
Unregistered user
Standard user
Premium user
Car
Supervisor
Time

Manager
Traffic information provider
Payment provider
Manage employees
AddAddress
System
Children

Name	Description
Register	User submits username, email, password, and payment card details on the registration page, the system validates the email format, password requirements, and payment card validity, checks if the email is already registered, sends a confirmation link to the user, verifies if the link is valid and confirmed, hashes the password, creates the user account in the database, and sends a welcome email.
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	User gets the option to change their requirements

Order ride	Description:
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	Primary actors: Registered User
	Secondary actors: Payment provider
	Preconditions: The user searched for a ride and at least one ride was provided
	Main flow:
	Registered User chooses a ride from the list of available rides
	Registered User is asked to confirm the selected ride
	if Registered User cancels the selection{
	Order ride()}
	else{
	if User's card details aren't saved in the system{
	Registered User is asked to fill in card information}
	else{
	Payment provider Processes payment
	if payment is successful{
	Car sets availability to unavailable
	System calculates route
	System sends route to car}}}
	Post-conditions:
	Car is unavailable, Firm received money
	Alternative flow:
	Payment fails:
	User receives an option to retry payment or cancel ride
	User logs out before payment:
	Action gets canceled

Subscribe to premium membership	User subscribes to premium membership using one of the approved payment methods, the system applies a discount if it applies checks if the payment was successful and sets the users account status to premium.
Create asset	The manager adds a new asset to the system by entering its relevant details. The new asset is saved into the system.
Edit asset	The manager modifies the details of an existing asset. The system saves the changes to ensure the asset information remains current.

Delete asset	The manager removes an asset that is no longer needed or available. The system updates its records to reflect the removal.
View statistics	The manager can access statistical information about system operations, users, assets and their usage.
Report arrival	The car's system reports that it has arrived at the pickup location. The system records the arrival information so that the route status can be updated.
Set availability	The car sets its status as available when it isn't currently driving passengers or en route to a pickup location or when its battery is above 40% and as unavailable otherwise.
Report technical issue	The car reports a technical problem or malfunction. In response, the system modifies the route or finds a suitable place to park if needed.
Report location	The vehicle sends its position to the system every second.
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Cancel route	If the passengers don't get onboard the vehicle within 15 minutes after it arrives to the pickup point, the system cancels the order.
Send refund	The system sends back the money for the canceled order after deducting a cancellation fee.
Save data	The system saves all data (orders, car information, routes, users, etc.) every second.

Process payment	The system processes payment for a ride or a premium membership. It communicates with the external payment provider to handle the transaction and determine its outcome.
Find a suitable place to park	When the car isn't carrying passengers and is available, the system finds a place to park and navigates the car there.
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Unsubscribe from premium membership	A premium user can cancel their premium subscription. The system updates the user's membership status and deletes their access to premium features.

Manage assets	Description:
	Manager gets the option to create, edit, or delete assets.
	Primary actors: Manager
	Secondary actors: -
	Preconditions: -
	Main flow:
	Manager opens the asset management section of the system
	Manager chooses to create new asset or selects an asset to edit/delete
	if Manager chooses to create new asset{
	Create asset()}
	if Manager selects an existing asset{
	if Manager chooses to edit the asset{
	Edit asset()}
	if Manager chooses to delete the asset{
	if Manager confirms the choice{
	Delete asset()}
	else{
	back to home screen}}
	System saves the updated assets}
	Post-conditions:
	One of the extensions is called
	Alternative flow:
	Manager logs out:
	Action gets canceled

AddAddress	
Manage employees	

Manage assets

Extension Points

ExtensionPoint	
Backlog Activity Id	0

ExtensionPoint	
Backlog Activity Id	0

ExtensionPoint	
Backlog Activity Id	0

Unsubscribe from premium membership

Provide state of the traffic

Review order history

Find a suitable place to park

Process payment

Send refund

Cancel route

Save data

Calculate route

Send notifications

Choose car accessories

Report battery status

Report location
Report technical issue
Set availability
Report arrival
View statistics
Delete asset
Edit asset
Create asset
Subscribe to premium membership
Sub Diagrams

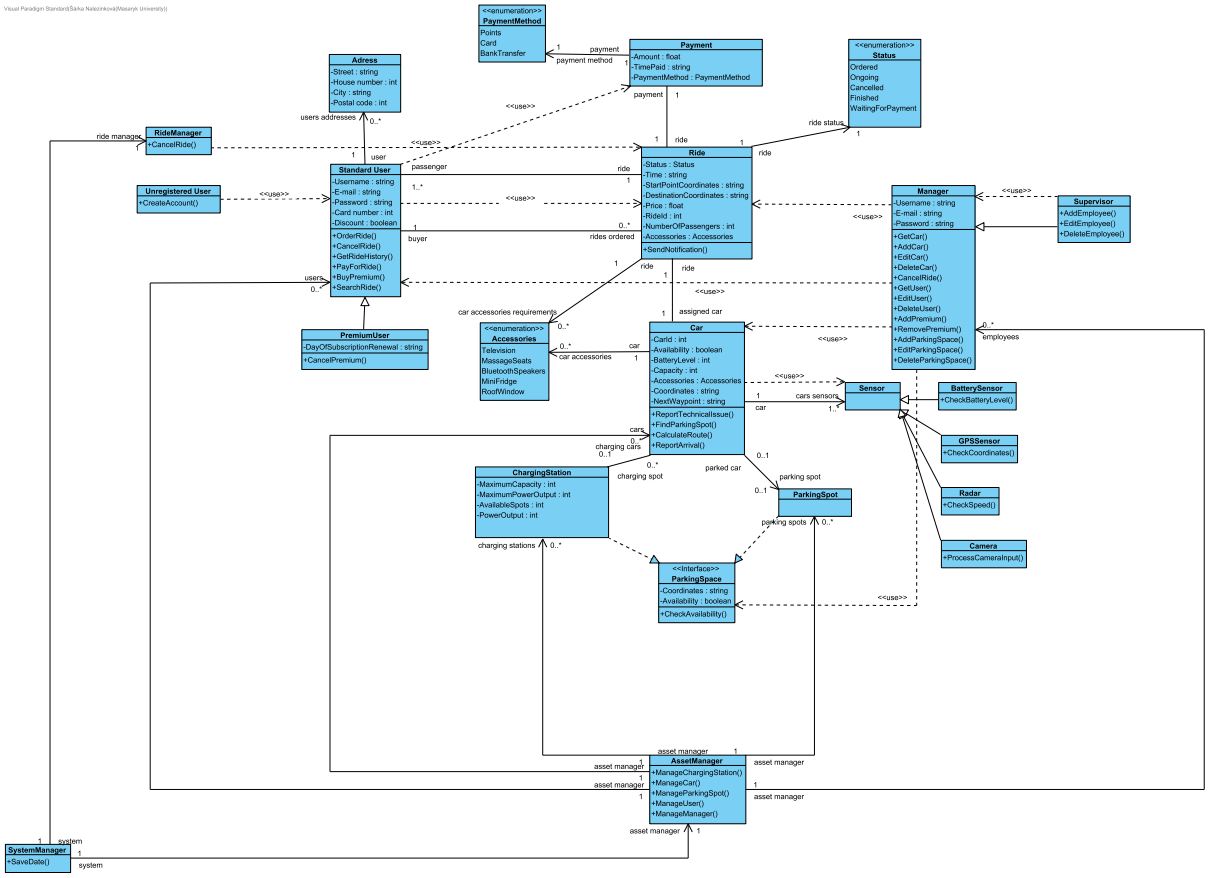
Name	Description
Subscribe to premium membership	

Order ride
Search for available cars
Register
Sub Diagrams

Name	Description
Register	

Class Diagram

Class Diagram1



Summary

Name	Description
PaymentMethod	
Payment	
Status	
Adress	
RideManager	
Ride	
Standard User	
Unregistered User	
Manager	
Supervisor	
Car	
Accessories	
PremiumUser	
Sensor	
BatterySensor	
GPSSensor	

ChargingStation	
Radar	
ParkingSpot	
Camera	
ParkingSpace	
AssetManager	
SystemManager	

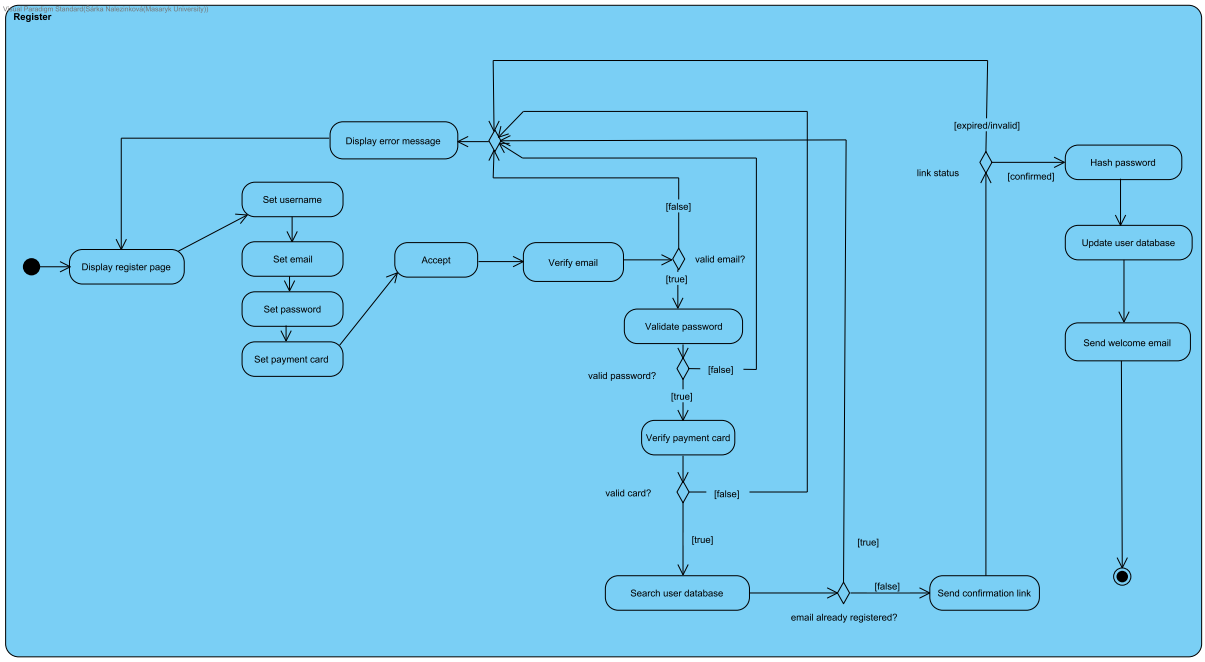
Details

PaymentMethod
Payment
Status
Adress
RideManager
Ride
Standard User
Unregistered User
Manager
Supervisor
Car
Accessories
PremiumUser
Sensor
BatterySensor
GPSSensor
ChargingStation
Radar
ParkingSpot
Camera
ParkingSpace
AssetManager

SystemManager

Activity Diagram

Register



Summary

Name	Description
Register	
Display register page	
Set username	
Set email	
Set password	
Set payment card	
Accept	
Verify email	
valid email?	
Validate password	
valid password?	
Verify payment card	
valid card?	
Search user database	
email already registered?	
Send confirmation link	
link status	
Hash password	
Update user database	

Send welcome email	
Display error message	

Details

N/A

Register

Children

Name	Description
N/A	
N/A	
Display register page	
Set username	
Set email	
Set password	
Accept	
email already registered?	
Display error message	
Send confirmation link	
link status	
N/A	
valid email?	
Verify email	
Search user database	
Update user database	
Set payment card	
Validate password	
valid password?	
Hash password	
Send welcome email	
Verify payment card	
valid card?	

Display register page

Set username

Set email

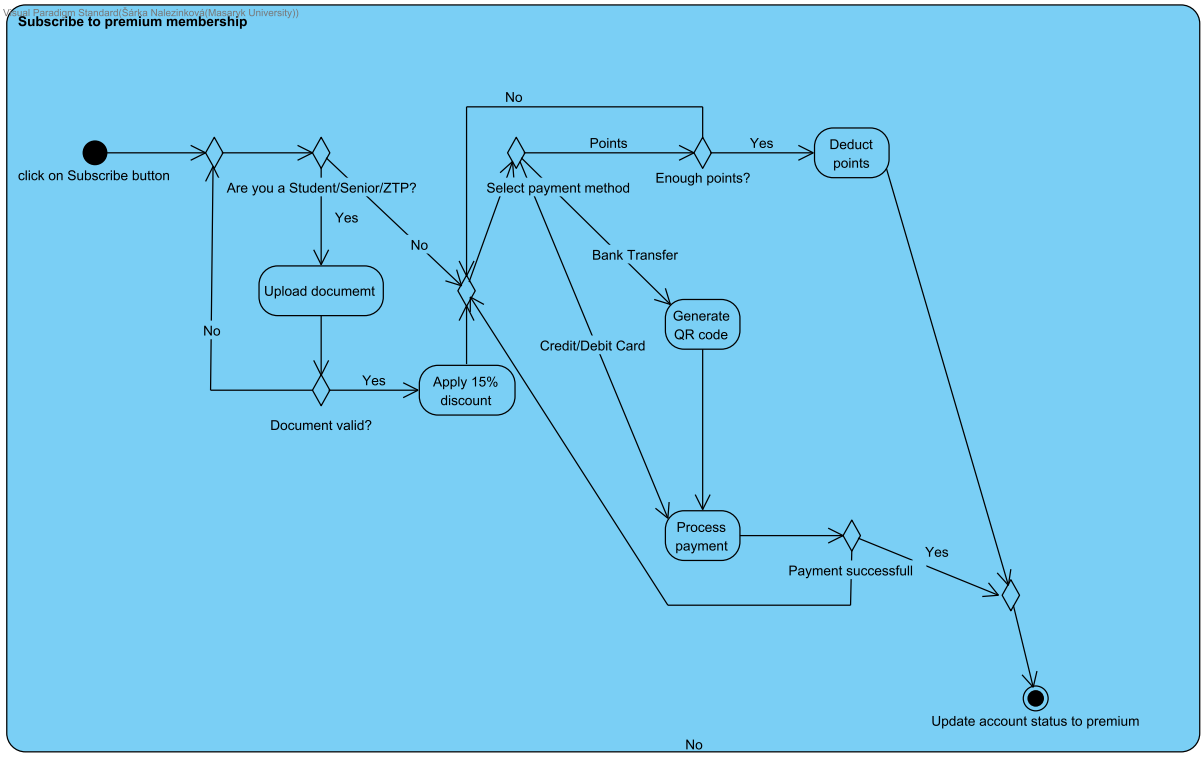
Set password

Set payment card

Accept
Verify email
valid email?
Validate password
valid password?
Verify payment card
valid card?
Search user database
email already registered?
Send confirmation link
link status
Hash password
Update user database
Send welcome email
N/A
N/A
Display error message

Activity Diagram

Subscribe to premium membership



Summary

Name	Description
click on Subscribe button	
Subscribe to premium membership	
Update account status to premium	
Payment successfull	
Generate QR code	
Deduct points	
Enough points?	
Process payment	
Select payment method	
Apply 15% discount	
Document valid?	
Upload documemt	
Are you a Student/Senior/ZTP?	

Details

click on Subscribe button

Subscribe to premium membership

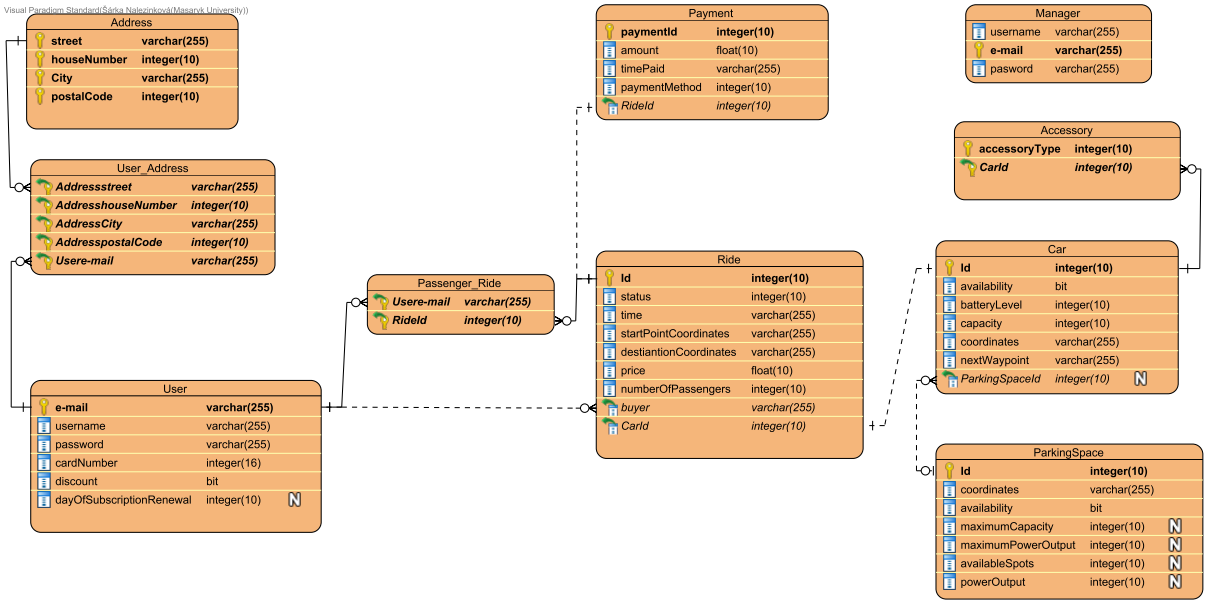
Children

Name	Description
click on Subscribe button	
Are you a Student/Senior/ZTP?	
Upload documemt	
Document valid?	
Apply 15% discount	
Select payment method	
Process payment	
Enough points?	
Deduct points	
Generate QR code	
Payment successfull	
N/A	
Update account status to premium	

Update account status to premium
N/A
Payment successfull
Generate QR code
Deduct points
Enough points?
Process payment
Select payment method
Apply 15% discount
Document valid?
Upload documemt
Are you a Student/Senior/ZTP?

Entity Relationship Diagram

Entity Relationship Diagram1



Summary

Name	Description
Payment	
Manager	
Address	
Accessory	
User_Address	
Car	
Ride	
Passenger_Ride	
User	
ParkingSpace	

Details

Payment

Manager

Address

Accessory

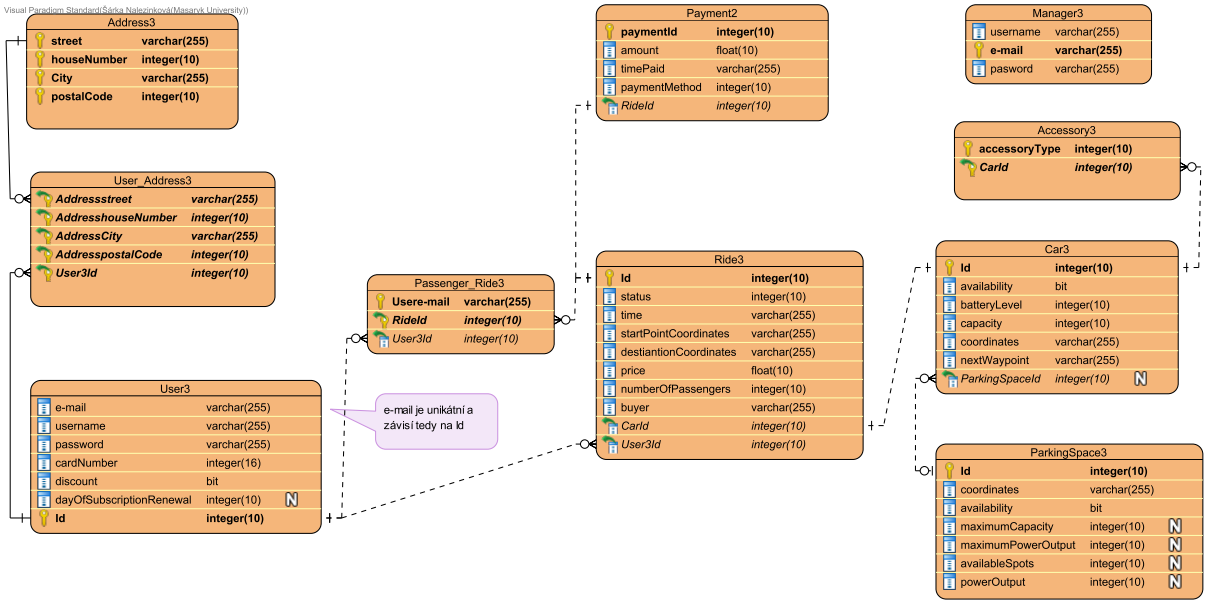
User_Address

Car

Ride
Passenger_Ride
User
ParkingSpace

Entity Relationship Diagram

Porušení 2 NF



Summary

Name	Description
Payment2	
Manager3	
Address3	
Accessory3	
User_Address3	
Car3	
Ride3	
Passenger_Ride3	
User3	
N/A	e-mail je unikátní a závisí tedy na Id
ParkingSpace3	

Details

Payment2

Manager3

Address3

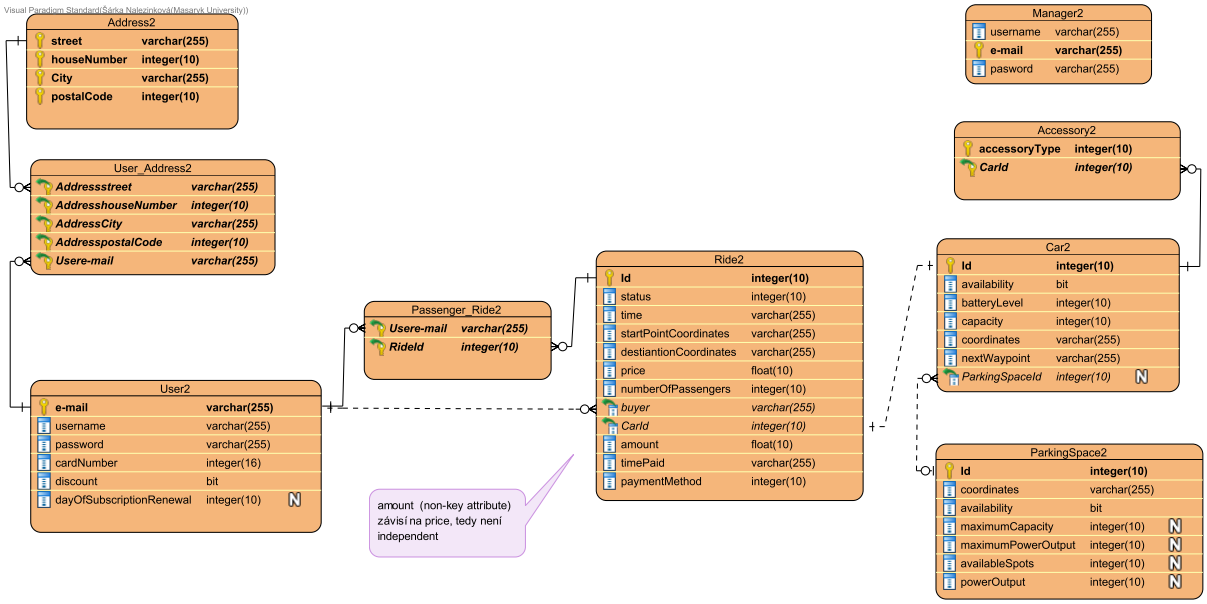
Accessory3

User_Address3

Car3
Ride3
Passenger_Ride3
User3
N/A
ParkingSpace3

Entity Relationship Diagram

Porušení 3 NF



Summary

Name	Description
Manager2	
Address2	
Accessory2	
User_Address2	
Car2	
Ride2	
Passenger_Ride2	
User2	
ParkingSpace2	
N/A	amount (non-key attribute) závisí na price, tedy není independent

Details

Manager2

Address2

Accessory2

User_Address2

Car2

Ride2

Passenger_Ride2
User2
ParkingSpace2
N/A